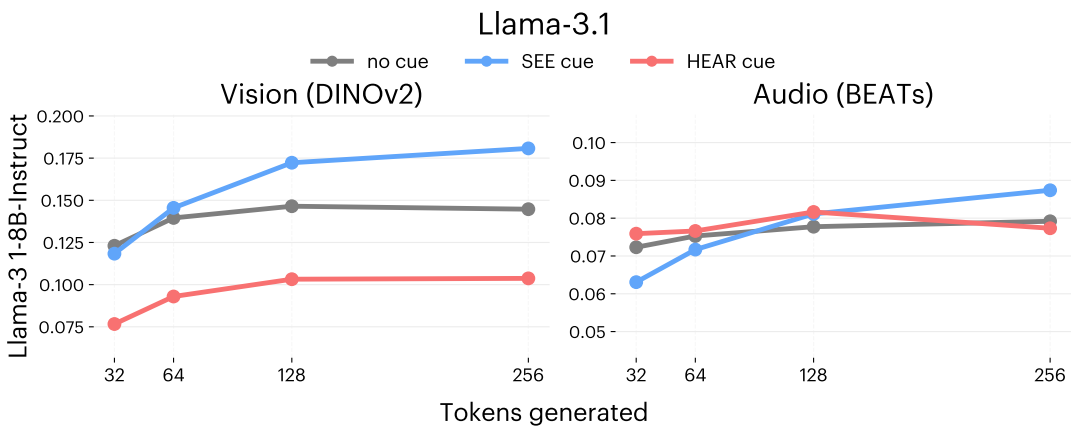

Figure 23: Extension of Figure 6 to additional language models: Llama 3.1.

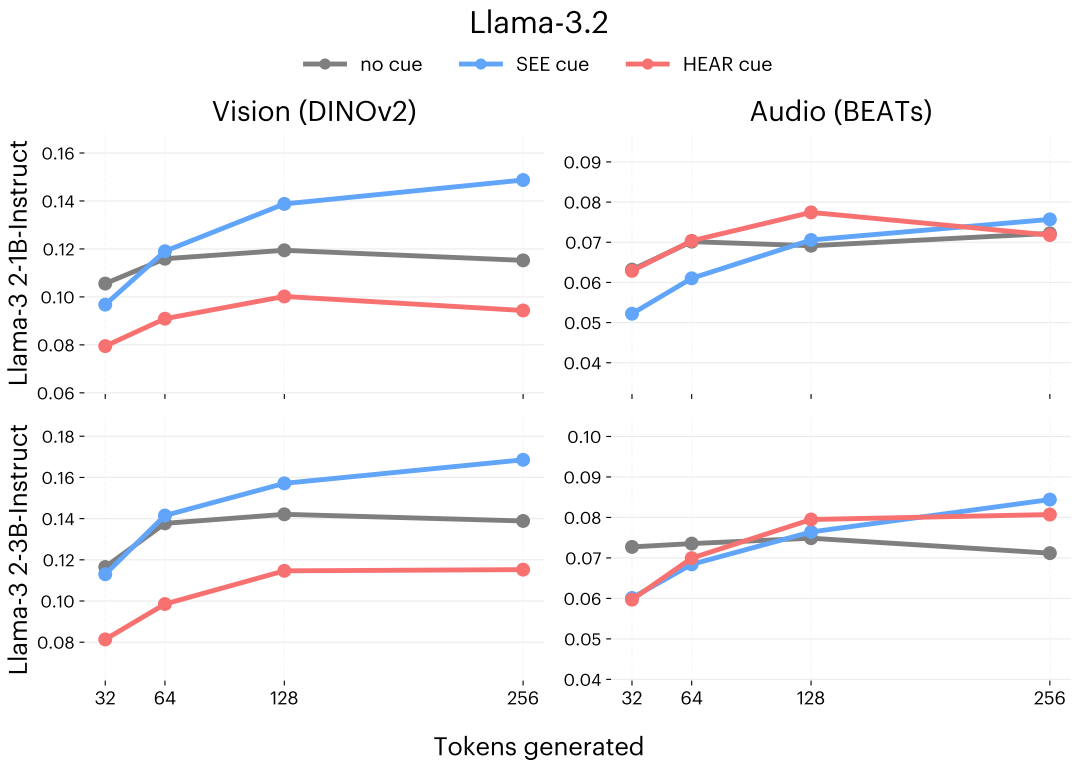

Figure 24: Extension of Figure 6 to additional language models: Llama 3.2.

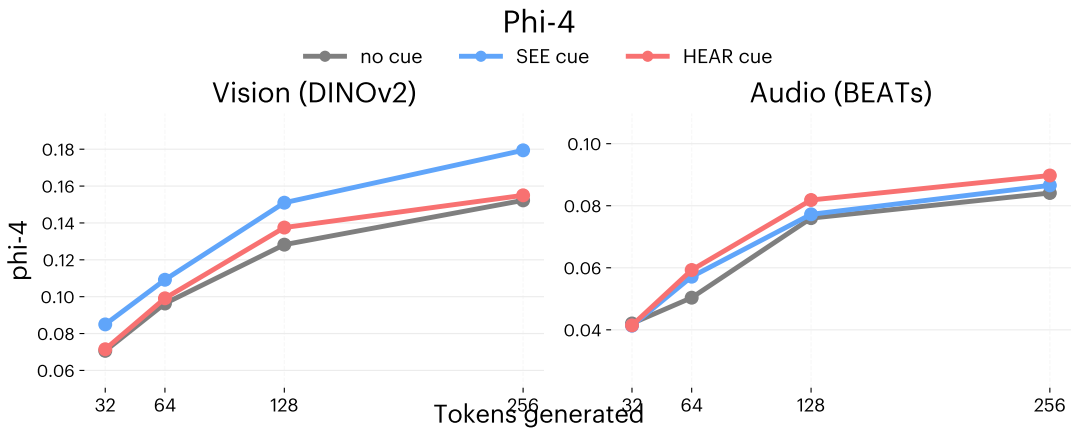

Figure 25: Extension of Figure 6 to additional language models: Phi-4.

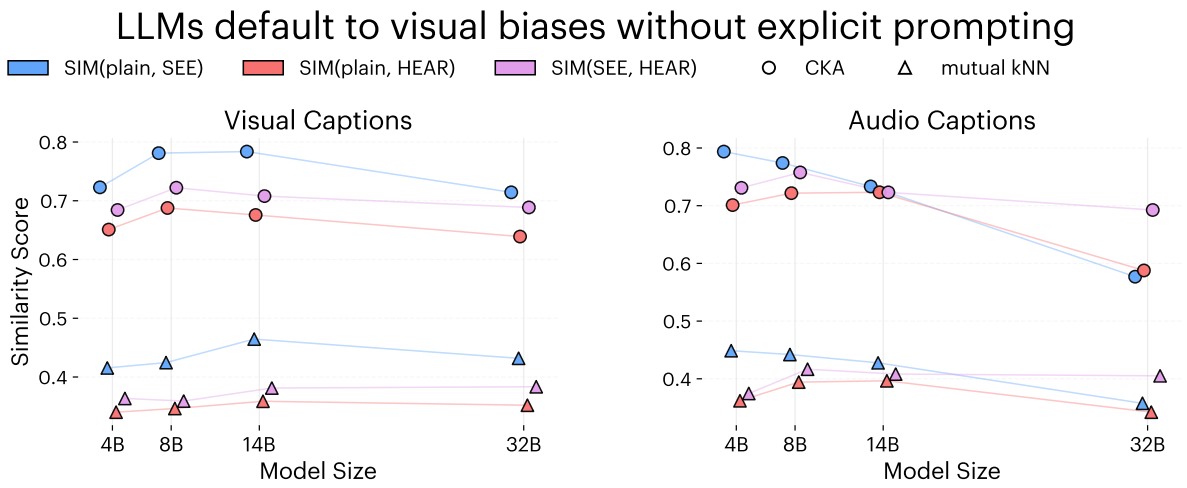

Figure 26: Similarity metrics (CKA and mutual-kNN) show that no cue prompts are consistently closer to SEE than HEAR—especially for audio captions—highlighting a default visual bias that diminishes with model scale.

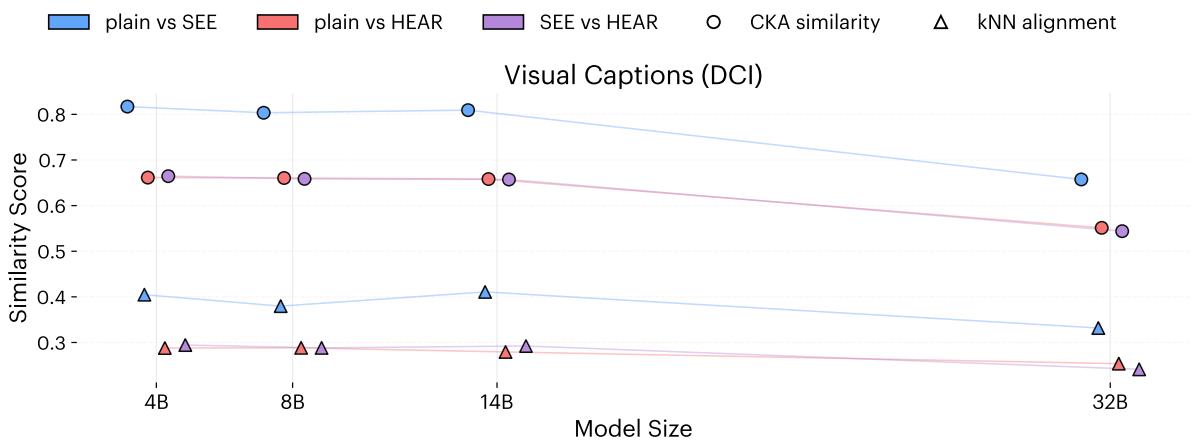

Figure 27: Extension of Figure 26 to "describe" instructed prompting.

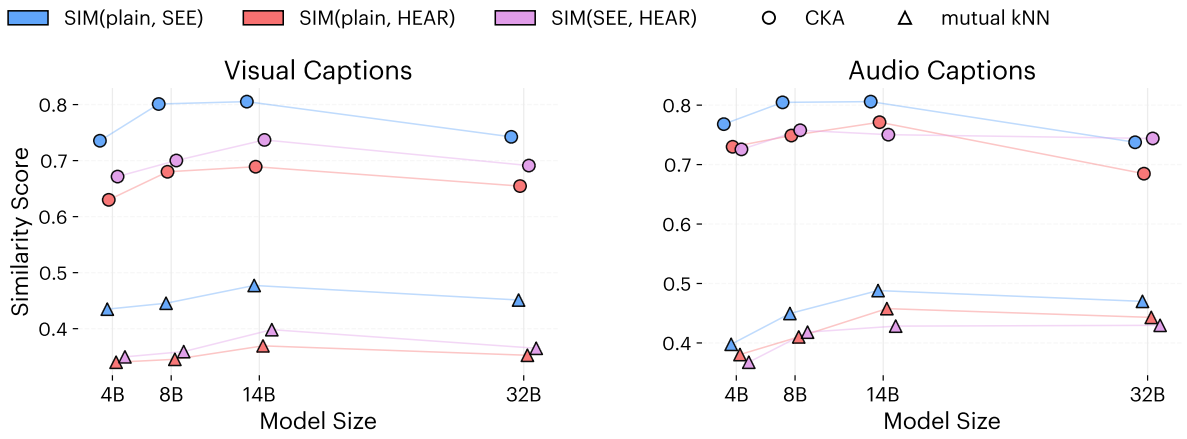

Figure 28: Extension of Figure 26 to the DCI dataset.

(Top). We observe that plain prompts are more evenly distributed between SEE and HEAR under the "describe" framing, suggesting that "describe" is more modality-neutral than "imagine," which may bias the model toward visual generation by default. This supports the idea that the instruction itself can influence how the model commits to a latent sensory framing, even in the absence of an explicit SEE or HEAR cue.

Table 2: Quantification of the visual–auditory disentanglement in Figure 8b.

Dataset	Model	$\Delta\mu$	Cohen's d	AUROC
WiT	Qwen3 4B	6.6	1.95	0.92
WiT	Qwen3 8B	13.3	2.13	0.94
WiT	Qwen3 14B	19.0	2.34	0.96
WiT	Qwen3 32B	21.0	2.65	0.97
AudioCaps	Qwen3 4B	10.6	3.14	0.98
AudioCaps	Qwen3 8B	19.4	3.10	0.98
AudioCaps	Qwen3 14B	28.8	4.11	0.99
AudioCaps	Qwen3 32B	32.4	4.52	1.00

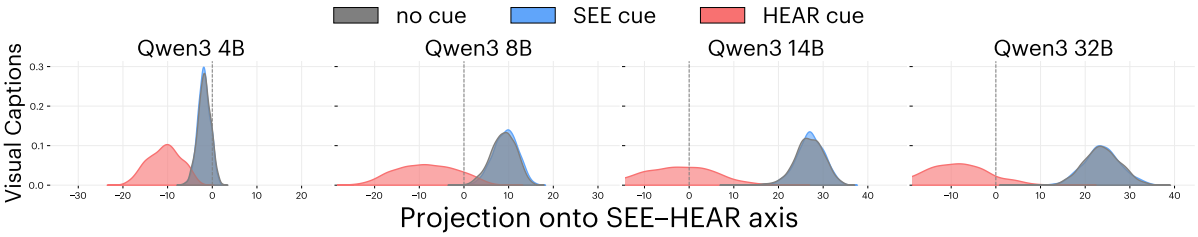

Figure 29: Extension of Figure 8 (Top) to DCI dataset.

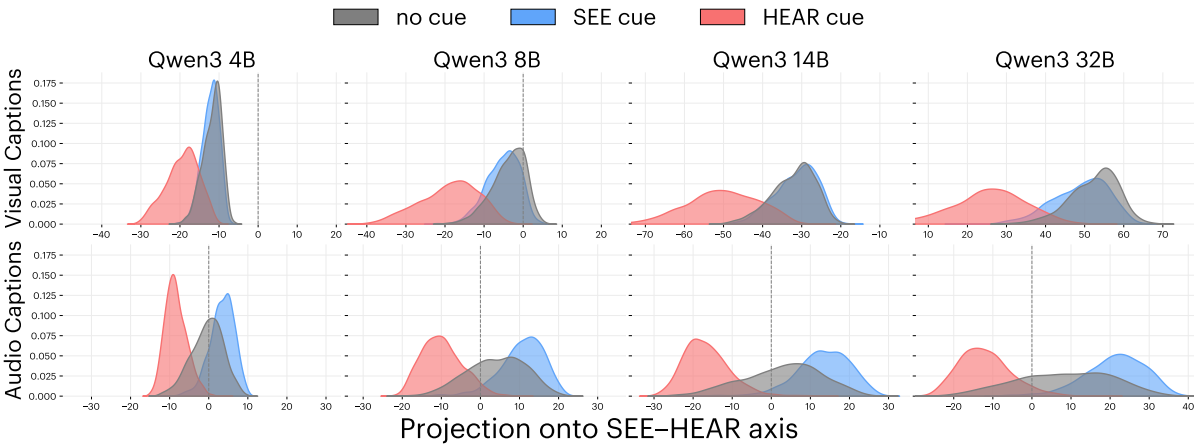

Figure 30: Extension of Figure 8 (Top) to "describe" instructed prompting.